Tugas Otomasi Bangunan

Prediksi Status Pemadaman Api di Bangunan Berbasis Gelombang Akustika

Menggunakan Algoritma Adaboost Classifier

M. Nadhir Al Ghifary

22/503570/TK/55030

Link Dataset: https://www.kaggle.com/datasets/muratkokludataset/acoustic-extinguisher-fire-dataset

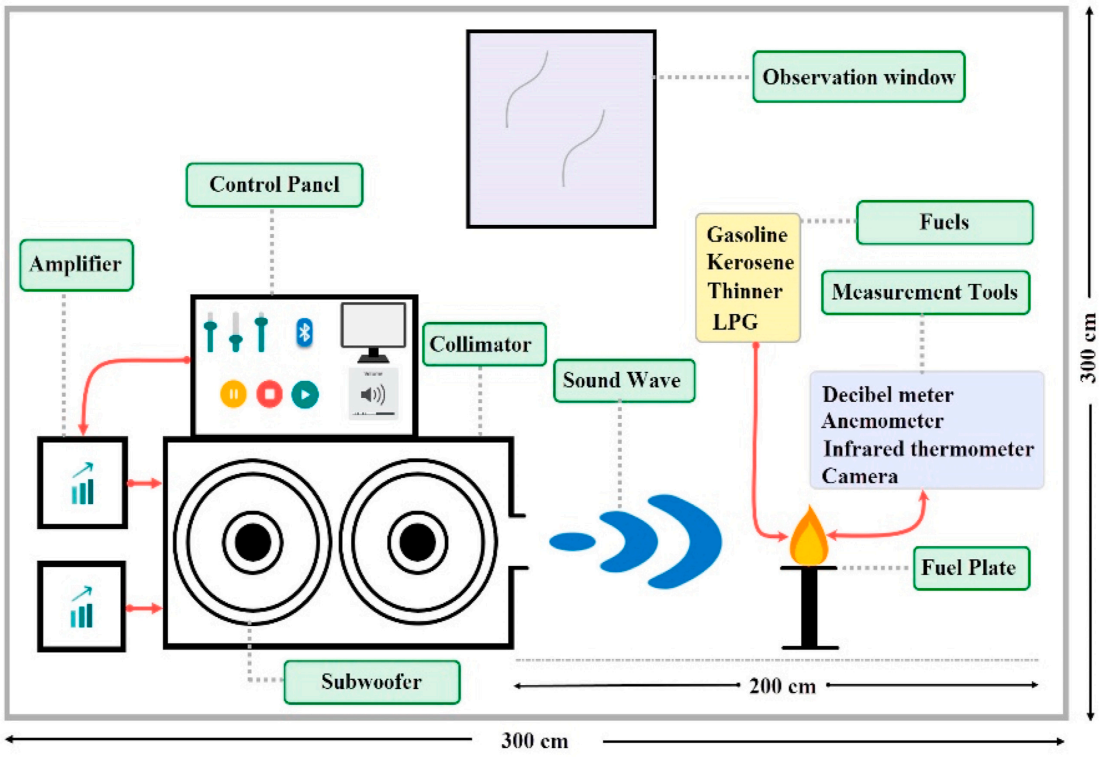
In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from datetime import timedelta
```

In [3]:
```
df = pd.read_excel("Acoustic_Extinguisher_Fire_Dataset.xlsx")
```

Set Up Eksperimen

Sebuah setup eksperimental dibuat untuk menguji pemadaman api dengan gelombang suara. Beberapa poin penting dalam setup eksperimen

- Setup eksperimental dibuat untuk mengeksplorasi konsep pemadaman api dengan menggunakan gelombang suara.
- Sistem pemadaman api dengan gelombang suara ditempatkan di dalam ruangan api yang terbuat dari bahan tahan api A1 yang non-mudah terbakar.
- Sistem ini terdiri dari kolimator dengan 4 subwoofer yang memiliki total daya sebesar 1000W (masing-masing subwoofer memiliki daya 250W).
- Terdapat juga 2 amplifier yang digunakan untuk memperkuat sinyal audio yang berasal dari generator sinyal. Setiap amplifier terhubung ke 2 subwoofer.
- Saat sistem dihidupkan, sistem mengambil arus tinggi secara instan, sehingga memerlukan pasokan daya khusus yang dirancang khusus.
- Kolimator diciptakan untuk memfokuskan aliran udara yang dihasilkan oleh gelombang suara pada satu titik target.
- Gelombang suara yang dihasilkan oleh getaran subwoofer dikirim melalui lubang berdiameter 13 cm pada kolimator untuk memengaruhi api.
- Kolimator memungkinkan gelombang suara untuk berkembang dari sumber ke target tanpa terdispersi.



#1: Telaah Data

A. Menampilkan Informasi Dataset

In [4]:
```
# Menampilkan Informasi Dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17442 entries, 0 to 17441
Data columns (total 7 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   SIZE       17442 non-null  int64
 1   FUEL       17442 non-null  object
 2   DISTANCE   17442 non-null  int64
 3   DESIBEL    17442 non-null  int64
 4   AIRFLOW    17442 non-null  float64
 5   FREQUENCY  17442 non-null  int64
 6   STATUS     17442 non-null  int64
dtypes: float64(1), int64(5), object(1)
memory usage: 954.0+ KB
```

B. Menampilkan Dataset

In [5]:
```
#Menampilkan 5 data awal dataset
df.head(10)
```

Out[5]:

	SIZE	FUEL	DISTANCE	DESIBEL	AIRFLOW	FREQUENCY	STATUS
0	1	gasoline	10	96	0.0	75	0
1	1	gasoline	10	96	0.0	72	1
2	1	gasoline	10	96	2.6	70	1
3	1	gasoline	10	96	3.2	68	1
4	1	gasoline	10	109	4.5	67	1
5	1	gasoline	10	109	7.8	66	1
6	1	gasoline	10	103	9.7	65	1
7	1	gasoline	10	95	12.0	60	1
8	1	gasoline	10	102	13.3	55	1
9	1	gasoline	10	93	15.4	52	1

In [6]:

```
#Menampilkan 5 data terakhir dataset
df.tail(5)
```

Out[6]:

	SIZE	FUEL	DISTANCE	DESIBEL	AIRFLOW	FREQUENCY	STATUS
17437	7	lpg	190	86	2.2	5	0
17438	7	lpg	190	84	2.0	4	0
17439	7	lpg	190	80	1.5	3	0
17440	7	lpg	190	76	0.4	2	0
17441	7	lpg	190	72	0.0	1	0

Berdasarkan telaah data tersebut, diperlukan perbaikan data sebelum visualiasi dan analisis statistik karena:

- Terdapat satu fitur yaitu `FUEL` yang memiliki tipe data `object`

#2: Perbaikan Data

A. Merubah Tipe Variabel dari Object ke Integer

In [7]:

```
# Membuat dictionary parameter seluruh object agar integer
perubahan = {'gasoline': 1,
             'thinner': 2,
             'kerosene': 3,
             'lpg': 4}

# Merubah object sesuai dengan dictionary yang dibuat
df['FUEL'] = df['FUEL'].replace(perubahan)
```

B. Menampilkan Ulang Info Datasheet Setelah Perbaikan Tipe Data

In [8]:
```
# Menampilkan Informasi Singkat dari Dataset setelah perubahan tipe variabel
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 17442 entries, 0 to 17441
Data columns (total 7 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   SIZE       17442 non-null  int64
 1   FUEL       17442 non-null  int64
 2   DISTANCE   17442 non-null  int64
 3   DESIBEL    17442 non-null  int64
 4   AIRFLOW    17442 non-null  float64
 5   FREQUENCY  17442 non-null  int64
 6   STATUS     17442 non-null  int64
dtypes: float64(1), int64(6)
memory usage: 954.0 KB
```

B. Menampilkan Ulang Datasheet

In [9]:
```
# Menampilkan Datasheet Setelah Perubahan
df.head(5)
```

Out[9]:

	SIZE	FUEL	DISTANCE	DESIBEL	AIRFLOW	FREQUENCY	STATUS
0	1	1	10	96	0.0	75	0
1	1	1	10	96	0.0	72	1
2	1	1	10	96	2.6	70	1
3	1	1	10	96	3.2	68	1
4	1	1	10	109	4.5	67	1

Berdasarkan hasil perbaikan data tersebut, data sudah seluruhnya bertipe `numeric` sehingga dapat dianalisis secara statistik

---

#3: Analisis Statistik Data

A. Menguji Kekosongan Data

In [10]:
```
# Menghitung persentase hilang tiap parameter dataset
persentase_datahilang = (df.isnull().sum() / len(df)) * 100

# Menampilkan persentae data hilang tiap parameter dataset
print('Persentase Data Hilang')
persentase_datahilang
```

```
Persentase Data Hilang
```

Out[10]:
```
SIZE         0.0
FUEL         0.0
DISTANCE     0.0
DESIBEL      0.0
AIRFLOW      0.0
FREQUENCY    0.0
STATUS       0.0
dtype: float64
```

B. Deskrispsi Statistik

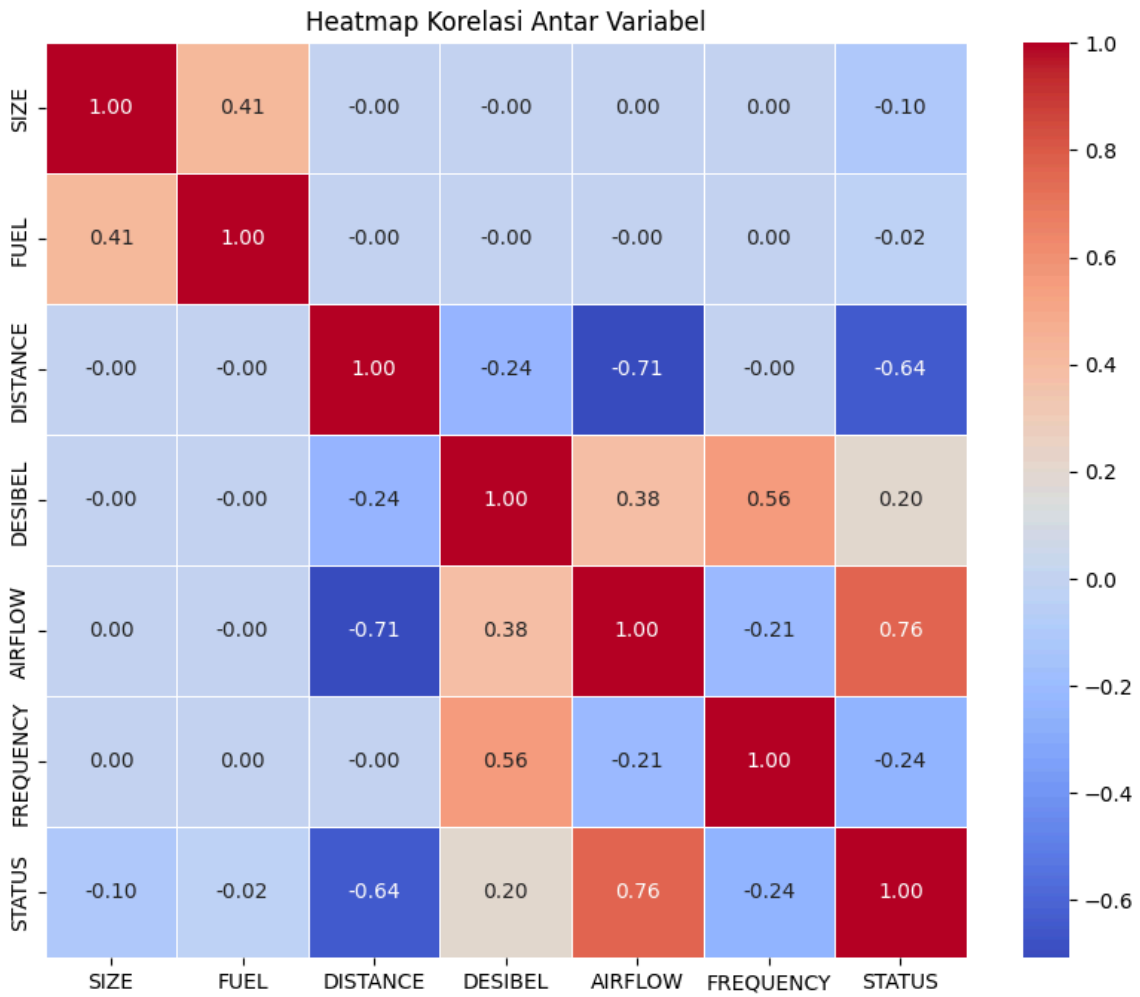
In [11]:
```
# Mengeksplorasi data
df.describe()
```

Out[11]:

	SIZE	FUEL	DISTANCE	DESIBEL	AIRFLOW	FREQUENCY	
count	17442.000000	17442.000000	17442.000000	17442.000000	17442.000000	17442.000000	17
mean	3.411765	2.235294	100.000000	96.379142	6.975634	31.611111	
std	1.750977	1.001757	54.773826	8.164096	4.736169	20.939149	
min	1.000000	1.000000	10.000000	72.000000	0.000000	1.000000	
25%	2.000000	1.000000	50.000000	90.000000	3.200000	14.000000	
50%	3.000000	2.000000	100.000000	95.000000	5.800000	27.500000	
75%	5.000000	3.000000	150.000000	104.000000	11.200000	47.000000	
max	7.000000	4.000000	190.000000	113.000000	17.000000	75.000000	

C. Korelasi Antar Variabel

In [12]:
```
# Hitung matriks korelasi
correlation_matrix = df.corr()

# Plot heatmap
plt.figure(figsize=(10, 8))  # Sesuaikan ukuran jika diperlukan
sns.heatmap(correlation_matrix, annot=True, cmap="coolwarm", fmt=".2f", linewidt
plt.title("Heatmap Korelasi Antar Variabel")
plt.show()
```

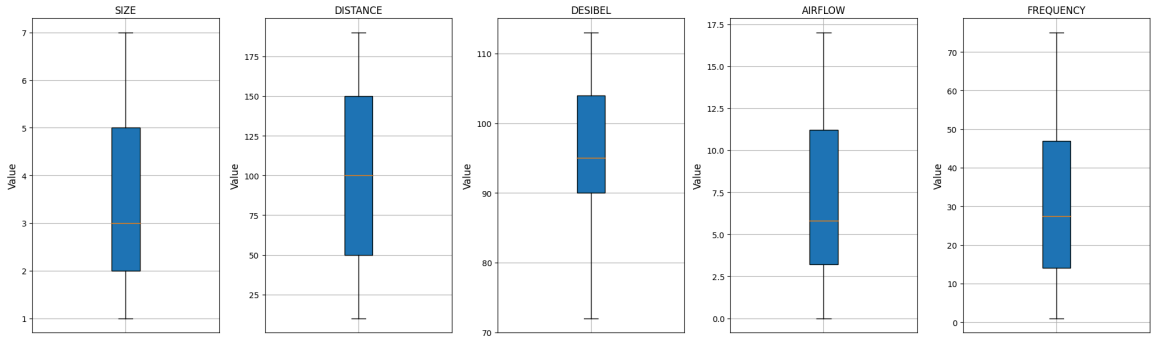


#4: Visualisasi Data

A. Boxplot Tiap Terukur

In [13]:
```
# Tentukan kolom untuk dikecualikan
columns_to_drop = ['FUEL', 'STATUS']

# Filter kolom kecuali yang dikecualikan
columns_to_plot = [col for col in df.columns if col not in columns_to_drop]

# Tentukan layout subplot (6 baris x 4 kolom)
num_cols = 5
num_rows = (len(columns_to_plot) + num_cols - 1) // num_cols  # Hitung jumlah ba

# Buat figure dan axes
fig, axes = plt.subplots(num_rows, num_cols, figsize=(20, 6))
axes = axes.flatten()  # Ratakan axes untuk iterasi lebih mudah

# Plot boxplot untuk setiap kolom yang dipilih
for i, col in enumerate(columns_to_plot):
    data = df[col].dropna()  # Hilangkan nilai NaN
    axes[i].boxplot(data, patch_artist=True)
    axes[i].set_title(col, fontsize=12)
    axes[i].set_ylabel("Value", fontsize=12)
    axes[i].grid(True)
    axes[i].tick_params(axis="x", which="both", bottom=False, labelbottom=False)
```

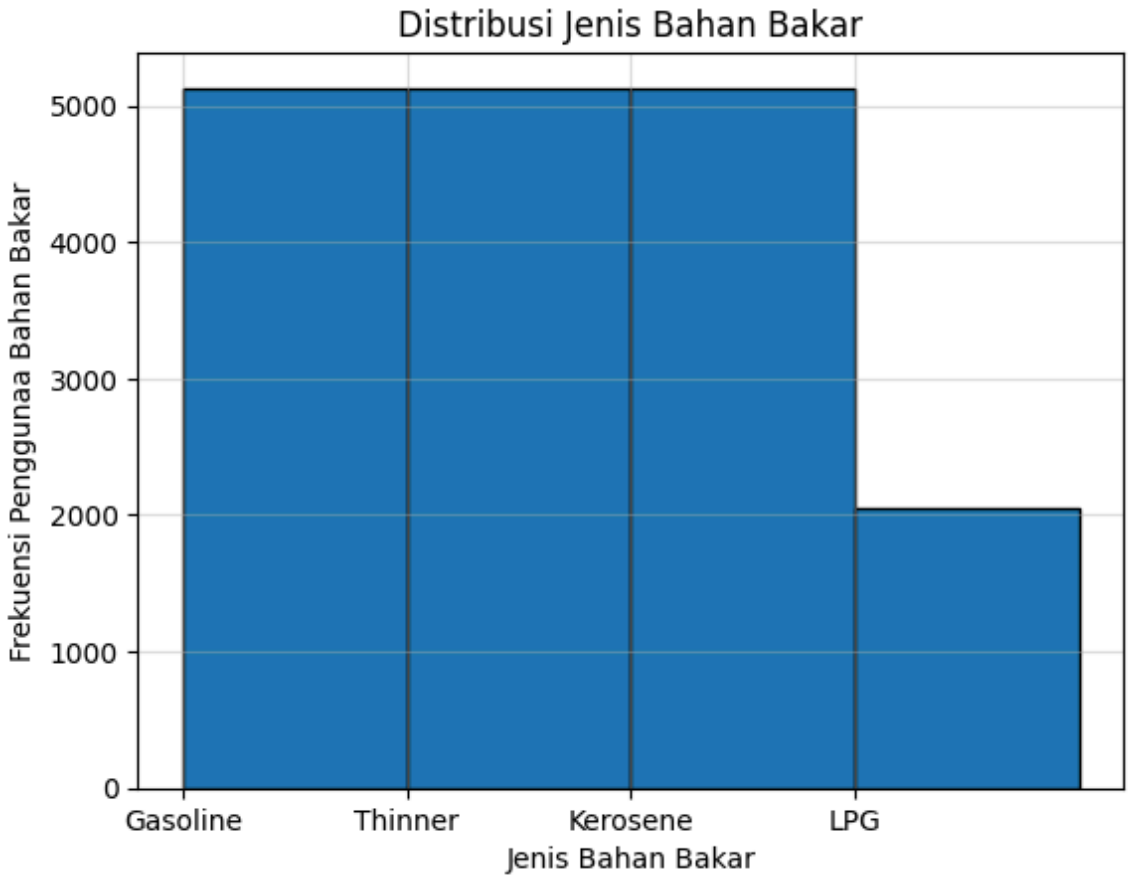
```
# Sembunyikan subplot kosong jika ada
for j in range(len(columns_to_plot), len(axes)):
    axes[j].axis('off')

# Atur tata letak subplot
plt.tight_layout()

# Tampilkan plot
plt.show()
```



Berdasarkan data ini, terlihat data sudah bersih dari outliers

B. Histogram Distribusi Jenis Bahan Bakar

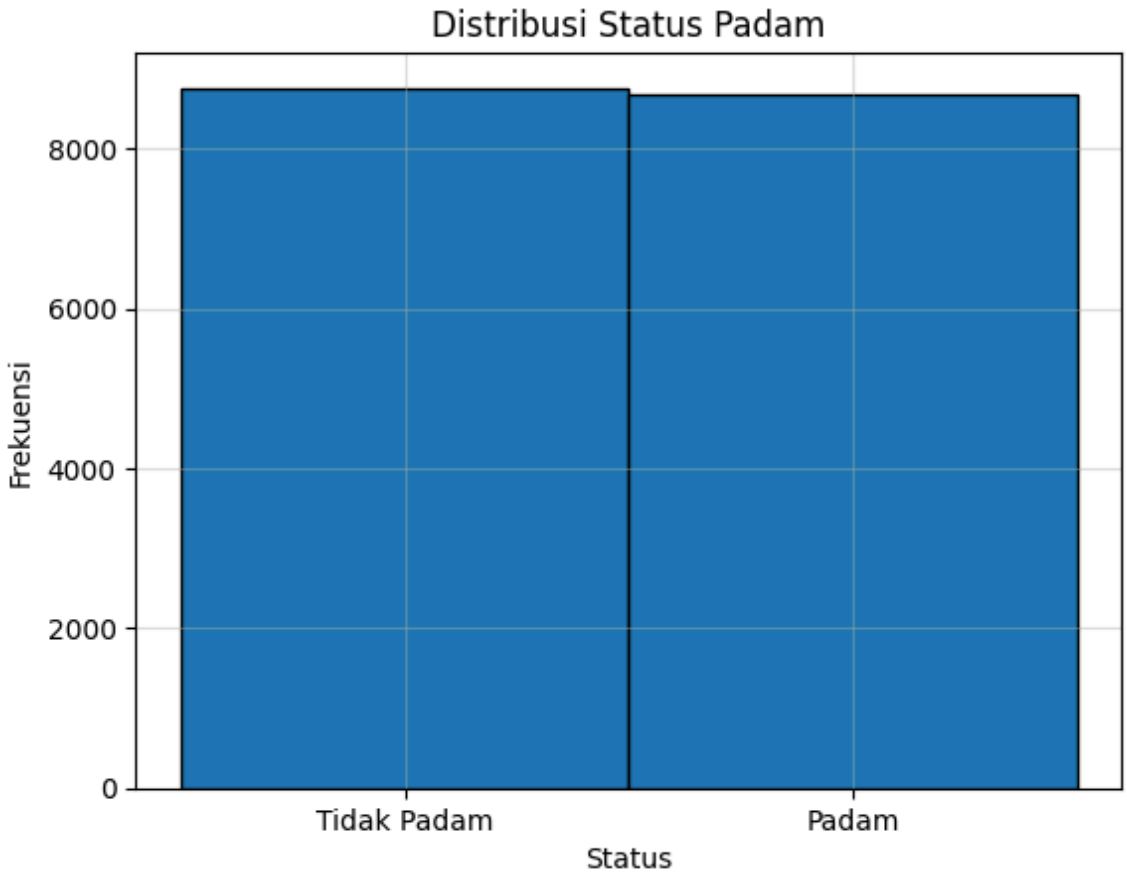
In [ ]:
```
# Membuat histogram
plt.hist(df['FUEL'], bins=range(1, 6), edgecolor='black')  # Pastikan bins sesua
plt.title('Distribusi Jenis Bahan Bakar')
plt.xlabel('Jenis Bahan Bakar')
plt.ylabel('Frekuensi Penggunaa Bahan Bakar')

# Mengubah keterangan di sumbu x
plt.xticks(ticks=[1, 2, 3, 4], labels=['Gasoline', 'Thinner', 'Kerosene', 'LPG']

plt.grid(alpha=0.4)
plt.show()
```



Berdasarkan histogram tersebut nampak data sudah termasuk seimbang. Gasoline, Thinner, dan Kerosene memiliki data yang sama. Adapun LPG lebih kecil secara kuantitas penggunaannya dari jenis bahan bakar lain

C. Distribusi Status Nyala Api

In [15]:
```
# Membuat histogram
plt.hist(df['STATUS'], bins=[-0.5, 0.5, 1.5], edgecolor='black', align='mid')  #
plt.title('Distribusi Status Padam')
plt.xlabel('Status')
plt.ylabel('Frekuensi')

# Mengubah keterangan di sumbu x
plt.xticks(ticks=[0, 1], labels=['Tidak Padam', 'Padam'])

# Menambahkan grid
plt.grid(alpha=0.4)
plt.show()
```



Berdasarkan distribusi tersebut nampak data yang seimbang antara Padam dan Tidak Padam sehingga sudah siap untuk dimodelkan

---

#5: Feature Engineering

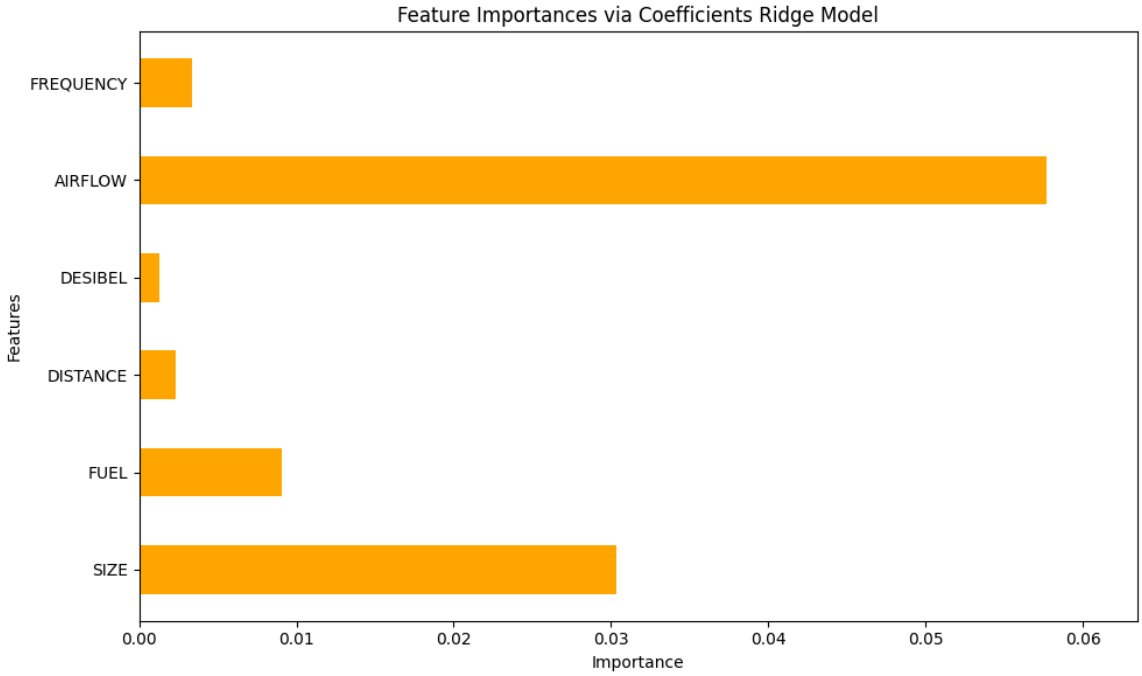
A. Membangi Data Train, Data Test, dan Data Validasi

In [16]:
```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import AdaBoostClassifier
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.tree import DecisionTreeClassifier

# Memisahkan fitur (X) dan target (y)
X = df.drop(columns=['STATUS'])  # Ganti dengan nama kolom target yang tepat
y = df['STATUS']

# Membagi dataset menjadi training dan testing set
X_train, X_temp, y_train, y_temp = train_test_split(X, y, test_size=0.3, random_
X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp, test_size=0.5, r
```

B. Feature Importance via Coefficient Ridge Model

In [17]:
```
from sklearn.linear_model import RidgeCV

# Model RidgeCV dengan alphas tertentu
ridge = RidgeCV(alphas=np.logspace(-3, 3, num=5)).fit(X_train, y_train)
```

```
# Menghitung feature importance berdasarkan absolut nilai koefisien
importance = np.abs(ridge.coef_)

# Nama fitur
if isinstance(X, pd.DataFrame):  # Jika X adalah DataFrame
    feature_names = X.columns
else:  # Jika X adalah array numpy
    feature_names = [f"Feature {i}" for i in range(X.shape[1])]

# Membuat plot dengan bar horizontal
plt.figure(figsize=(10, 6))
plt.barh(y=feature_names, width=importance, color='orange', height=0.5)
plt.title("Feature Importances via Coefficients Ridge Model")
plt.xlabel("Importance")
plt.ylabel("Features")
plt.xlim(0, max(importance) * 1.1)  # Menyesuaikan batas x agar proporsional
plt.tight_layout()
plt.show()
```



In [18]:
```
# Memisahkan fitur (X) dan target (y)
X = df.drop(columns=['STATUS'])  # Ganti dengan nama kolom target yang tepat
y = df['STATUS']
```

#6: Model Building → Adaboost Classifier - Decission Tree

A. Adaboost Classifier - Decission Tree

Train Model Metode Train Validation Split

In [36]:
```
weak_learner = DecisionTreeClassifier(max_leaf_nodes=5)
n_estimators = 100

adaboost_clf = AdaBoostClassifier(
    base_estimator=weak_learner,
    n_estimators=n_estimators,
```

```
    algorithm="SAMME",
    random_state=42,
)

adaboost_clf.fit(X_train, y_train)
```

Out[36]:

```
▸ AdaBoostClassifier
▸ base_estimator: DecisionTreeClassifier
    ▸ DecisionTreeClassifier
```

Analisis Konvergensi AdaBoostClassifier

In [20]:

```
from sklearn.dummy import DummyClassifier
from sklearn.metrics import accuracy_score

dummy_clf = DummyClassifier()

def misclassification_error(y_true, y_pred):
    return 1 - accuracy_score(y_true, y_pred)

weak_learners_misclassification_error = misclassification_error(
    y_test, weak_learner.fit(X_train, y_train).predict(X_test)
)

dummy_classifiers_misclassification_error = misclassification_error(
    y_test, dummy_clf.fit(X_train, y_train).predict(X_test)
)

print(
    "DecisionTreeClassifier's misclassification_error: "
    f"{weak_learners_misclassification_error:.3f}"
)
print(
    "DummyClassifier's misclassification_error: "
    f"{dummy_classifiers_misclassification_error:.3f}"
)
```

```
DecisionTreeClassifier's misclassification_error: 0.135
DummyClassifier's misclassification_error: 0.500
```

Setelah melatih DecisionTreeClassifiermodel, kesalahan yang dicapai melampaui nilai yang diharapkan yang akan diperoleh dengan menebak label kelas yang paling sering muncul, seperti yang DummyClassifierterjadi.

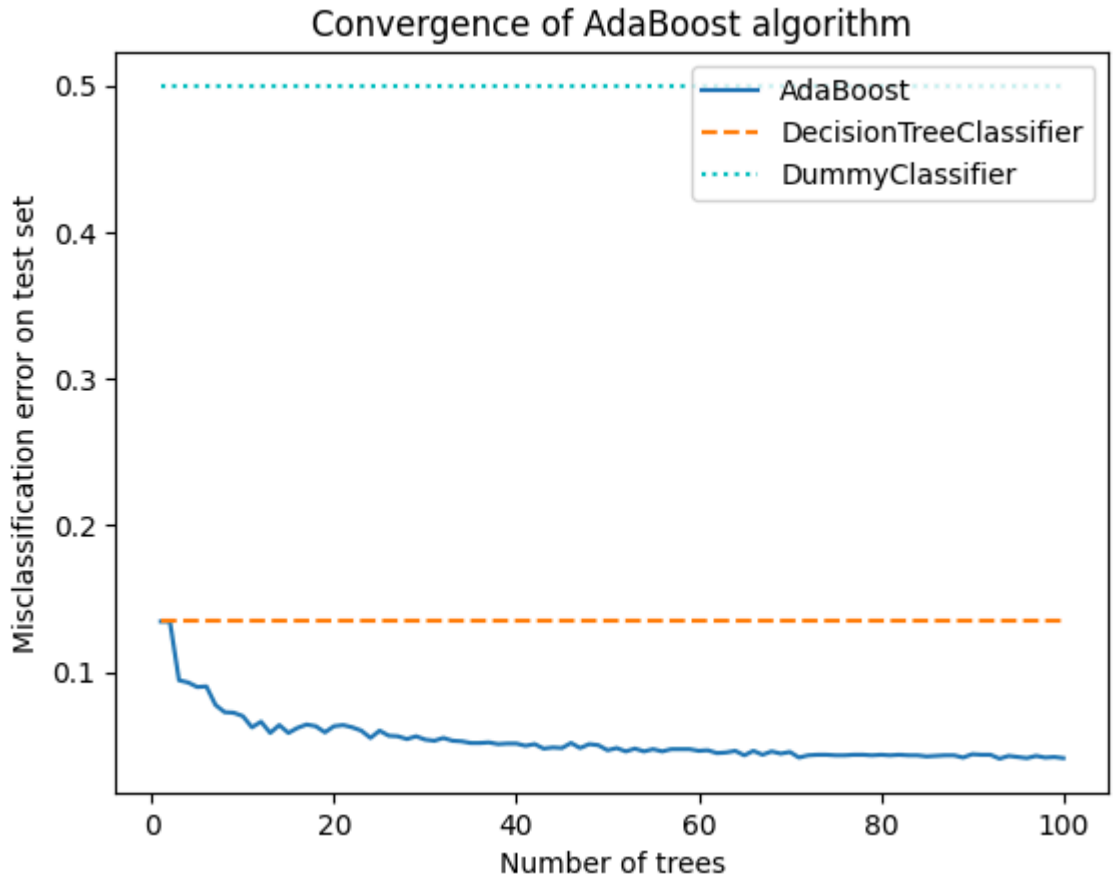
Staged Predict Error untuk Melakukan Boosting Setiap Iterasi Pemodelan

In [25]:

```
boosting_errors = pd.DataFrame(
    {
        "Number of trees": range(1, n_estimators + 1),
        "AdaBoost": [
            misclassification_error(y_test, y_pred)
            for y_pred in adaboost_clf.staged_predict(X_test)
        ],
    }
```

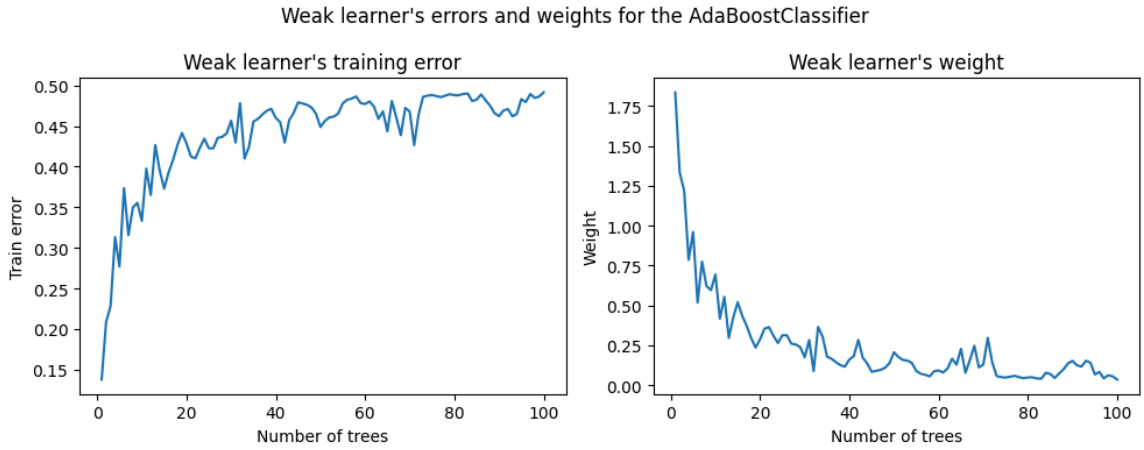
```
).set_index("Number of trees")
ax = boosting_errors.plot()
ax.set_ylabel("Misclassification error on test set")
ax.set_title("Convergence of AdaBoost algorithm")

plt.plot(
    [boosting_errors.index.min(), boosting_errors.index.max()],
    [weak_learners_misclassification_error, weak_learners_misclassification_erro
    color="tab:orange",
    linestyle="dashed",
)
plt.plot(
    [boosting_errors.index.min(), boosting_errors.index.max()],
    [
        dummy_classifiers_misclassification_error,
        dummy_classifiers_misclassification_error,
    ],
    color="c",
    linestyle="dotted",
)
plt.legend(["AdaBoost", "DecisionTreeClassifier", "DummyClassifier"], loc=1)
plt.show()
```

```
c:\Users\ACER\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn
\base.py:450: UserWarning: X does not have valid feature names, but AdaBoostCla
ssifier was fitted with feature names
  warnings.warn(
```



In [22]:
```
weak_learners_info = pd.DataFrame(
    {
        "Number of trees": range(1, n_estimators + 1),
        "Errors": adaboost_clf.estimator_errors_,
        "Weights": adaboost_clf.estimator_weights_,
```

```
    }
).set_index("Number of trees")

axs = weak_learners_info.plot(
    subplots=True, layout=(1, 2), figsize=(10, 4), legend=False, color="tab:blue
)
axs[0, 0].set_ylabel("Train error")
axs[0, 0].set_title("Weak learner's training error")
axs[0, 1].set_ylabel("Weight")
axs[0, 1].set_title("Weak learner's weight")
fig = axs[0, 0].get_figure()
fig.suptitle("Weak learner's errors and weights for the AdaBoostClassifier")
fig.tight_layout()
```



B. Random Forest Classifier

Parameter Pemodelan dan Training

In [29]:
```
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, f1_score, precision_score, recall_sc

rf_clf = RandomForestClassifier(
    n_estimators=100,      # mirip jumlah estimator AdaBoost
    max_depth=None,        # biarkan pohon full depth (default)
    random_state=42,
    n_jobs=-1              # gunakan semua core CPU agar lebih cepat
)

rf_clf.fit(X_train, y_train)
```

Out[29]:
```
▾ RandomForestClassifier
RandomForestClassifier(n_jobs=-1, random_state=42)
```

C. Gradient Boosting

In [ ]:
```
from sklearn.ensemble import GradientBoostingClassifier
from sklearn.metrics import accuracy_score, f1_score, precision_score, recall_sc

gb_clf = GradientBoostingClassifier(
    n_estimators=100,     # sama seperti AdaBoost dan Random Forest
    learning_rate=0.1,    # default, biasanya sudah optimal
    max_depth=3,          # sama seperti weak learner tree depth di boosting
```

```
    random_state=42
)

gb_clf.fit(X_train, y_train)
```

Out[ ]:

```
GradientBoostingClassifier
GradientBoostingClassifier(random_state=42)
```

#7: Evaluasi Model

Komparisi Performa Tiap Model

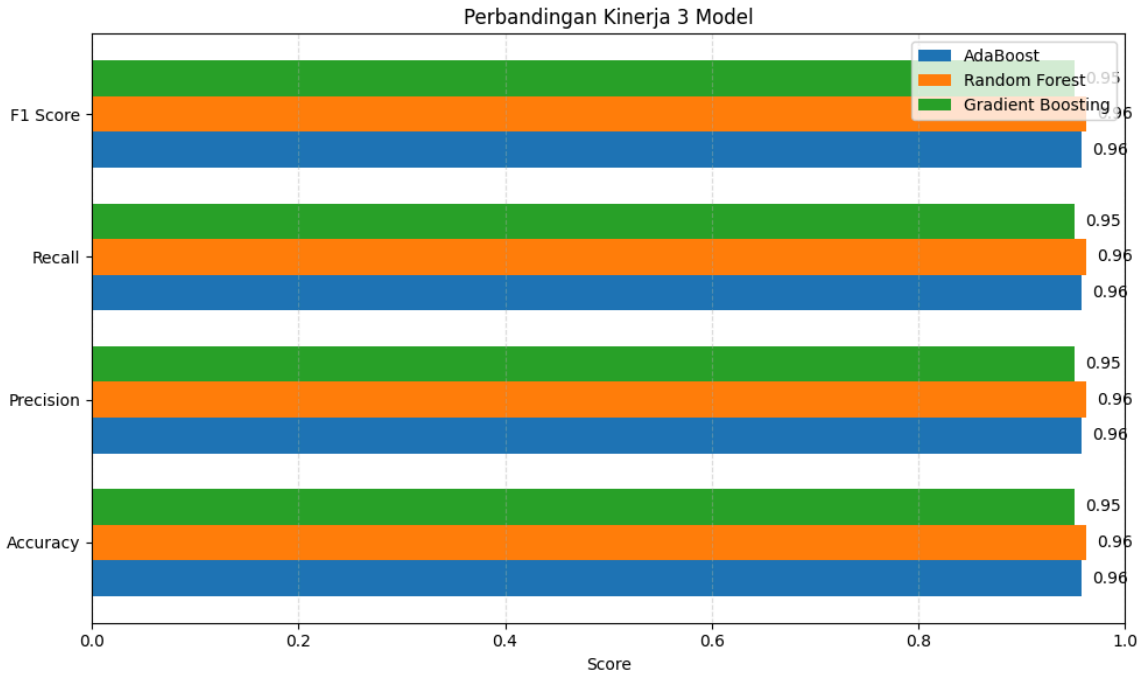
In [34]:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score, f1_score, precision_score, recall_sc

ada_pred = adaboost_clf.predict(X_test)
rf_pred = rf_clf.predict(X_test)
gb_pred = gb_clf.predict(X_test)

def get_metrics(y_true, y_pred):
    return {
        "Accuracy": accuracy_score(y_true, y_pred),
        "Precision": precision_score(y_true, y_pred, average="weighted"),
        "Recall": recall_score(y_true, y_pred, average="weighted"),
        "F1 Score": f1_score(y_true, y_pred, average="weighted"),
    }

ada_metrics = get_metrics(y_test, ada_pred)
rf_metrics = get_metrics(y_test, rf_pred)
gb_metrics = get_metrics(y_test, gb_pred)

# Visualisasi Classification Report

print("=== AdaBoost Report ===")
print(classification_report(y_test, ada_pred))

print("=== Random Forest Report ===")
print(classification_report(y_test, rf_pred))

print("=== Gradient Boosting Report ===")
print(classification_report(y_test, gb_pred))
```

```
=== AdaBoost Report ===
              precision    recall  f1-score   support

           0       0.96      0.95      0.96      1309
           1       0.96      0.96      0.96      1308

    accuracy                           0.96      2617
   macro avg       0.96      0.96      0.96      2617
weighted avg       0.96      0.96      0.96      2617

=== Random Forest Report ===
              precision    recall  f1-score   support

           0       0.97      0.95      0.96      1309
           1       0.96      0.97      0.96      1308

    accuracy                           0.96      2617
   macro avg       0.96      0.96      0.96      2617
weighted avg       0.96      0.96      0.96      2617

=== Gradient Boosting Report ===
              precision    recall  f1-score   support

           0       0.95      0.95      0.95      1309
           1       0.95      0.95      0.95      1308

    accuracy                           0.95      2617
   macro avg       0.95      0.95      0.95      2617
weighted avg       0.95      0.95      0.95      2617
```

Visualisasi Perbandingan Tiap Model

In [35]:

```
# Grafik Perbandingan (Matplotlib)

labels = list(ada_metrics.keys())  # ['Accuracy', 'Precision', 'Recall', 'F1 Sco

ada_values = list(ada_metrics.values())
rf_values = list(rf_metrics.values())
gb_values = list(gb_metrics.values())

# Membuat posisi Y
y = np.arange(len(labels))
height = 0.25  # ketebalan bar

plt.figure(figsize=(10, 6))

plt.barh(y - height, ada_values, height, label="AdaBoost")
plt.barh(y,          rf_values, height, label="Random Forest")
plt.barh(y + height, gb_values, height, label="Gradient Boosting")

plt.xlim(0, 1)
plt.xlabel("Score")
plt.title("Perbandingan Kinerja 3 Model")
plt.yticks(y, labels)
plt.grid(axis="x", linestyle="--", alpha=0.4)
plt.legend()

# Tambahkan nilai pada bar
```

```python
for i, v in enumerate(ada_values):
    plt.text(v + 0.01, i - height, f"{v:.2f}", va="center")

for i, v in enumerate(rf_values):
    plt.text(v + 0.01, i, f"{v:.2f}", va="center")

for i, v in enumerate(gb_values):
    plt.text(v + 0.01, i + height, f"{v:.2f}", va="center")

plt.tight_layout()
plt.show()
```